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Memory Game

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Introduction:

This is a software project for a memory game. The goal of this project is to allow older people with memory problems to play an entertaining game that can be one of many components to stave off short-term memory loss(Abd-alrazaq et al., 2023). The layout of the game will be a button memory game, where there will be a set amount of buttons, each unique, and they will play a sequence/order, and the user will have to repeat that order. Once the order has been successfully repeated, the buttons will play the same sequence that just occurred, but with an additional button click at the end.

There will be about six main functionalities of this software/game. The first and most important function is the actual game and its button sequence, which will have the user repeat back the computer-generated sequence of buttons, this is expanded on in the first paragraph of the introduction section(just above this paragraph). The next functionalities are not in any order of importance. First is the difficulty functionality. This will include a couple of different settings that will allow the user to alter the difficulty of the game, this will include the ability to change the number of pressable buttons in the game, this difficulty functionality also may include a goal mode, where a goal score has to be reached, this will be further talked about in the scoring function. There will also be a leaderboard section, which is a list of people who have the highest score. The fourth function is the scoring functionality. This would keep track of how many buttons have been pressed in the current session. The fifth functionality would be a colorblind section, which would allow the button colors to be changed. The sixth functionality would be the settings, which would include various changes to be made, which could include the text size, sound volume, and other settings that may be needed. The last functionality would be a help screen, which would the user understand how the game works.

There are quite a few different tools that will be used to fully develop this project. There are three main tools needed to actually create the website/application, which include HTML, CSS, JS, and PHP. There will also be the use of Git/Github in order to share the code and update the code(a version control system). FL Studio or a similar application will need to be used in order to develop sounds. There is also the alternative of acquiring copyright-free sounds off of the internet.

Potential users include first and foremost old people, which are the primary age of people who have memory issues, the latter of which this game is trying to combat. There could also be people who like to play games in general/puzzle games. or those who want to test their short-term memory skills.

Benefits could include keeping the brain sharp and active(Combs, 2023), entertainment and stress relief(Scott, 2024), and competition. Competition could be enhanced with the use of a leaderboard and may encourage motivation/competition with oneself.

Changes

The functionalities that have been updated from the original paper requirements are that there will not be a time or moves counter, and there also be a colorblind ability. We have also chosen the main game's goal. We have also chosen the fact that HTML/CSS/JS will be the software of choice, compared to Python or Java. There also will likely be no need to use graphics implementation, which can be done in one of the other preexisting languages. A note is this game is subject to change, with the possible removal and addition of some features which may be big or small.

There are also some other updates that have been made since the making of the first version of the system requirements assignment. This is the layout of the introduction, where the functionalities have been changed from a list into a full paragraph. One functionality to the game has been made, which is a help screen. Requirements have been modified to meet formal expectations. The tools used paragraph has also been expanded on. System requirements have also been updated.

Requirements Specification:

Functional Requirements

1. The software shall have a main menu screen with buttons that take users to other screens: start button, settings button, leaderboard button
2. The game shall have a start button to play the game.
3. The game shall display the sequence of buttons for the player to remember, which increases in length every round
4. The game shall have a sequence that is easily able to be replicated by the user
5. The game shall check if the user's repeated sequence is correct. If the user sequence is correct, the game shall go to the next round. If the user sequence is not correct, the game will end, resulting in an end-game screen
6. The game shall keep track of the score and a possible goal the user/computer has set
7. The game shall display the score at the end game screen, and if a goal has been met. The game shall also put the score on the leaderboard if the requirements are met

8. The leaderboard button shall display the leaderboard- which is (3 or 10) of the current high scores,
9. The restart button, displayed on the end game screen, shall restart the game to replay the game
10. The leaderboard button, displayed on the end game screen and main menu screen, shall properly display the leaderboard screen
11. The settings button, on the main menu screen and end game screen, shall access the settings screen

12. The settings screen shall properly allow editing of the text size, volume, colorblind mode, and difficulty settings- goal model or free play mode, and amount of buttons.
13. If the user is in goal-oriented mode, then once the user reaches the goal score, the goal screen should be displayed, and allow the user to restart, or possibly change the mode they are playing in some way
14. When the game is in goal mode, then the game shall stop when the user has reached that mode, and display similar buttons to that of the normal end game screen. The user shall be able to choose if they want to have another goal/progress.

Nonfunctional Requirements

1. The game shall have minimal delay, to less than one second when actually playing the game, and less than five seconds when switching between screens or settings
2. The game shall be able to be run on at least Microsoft Edge, Google Chrome, and Firefox.
3. The game shall be easy to follow, and the average user shall be able to play the game with no assistance within a five-minute training and experience period.
4. The game's website shall not use any more than 300 megabytes of RAM.
5. The game shall not crash more than once per 100 sessions.

System requirements:

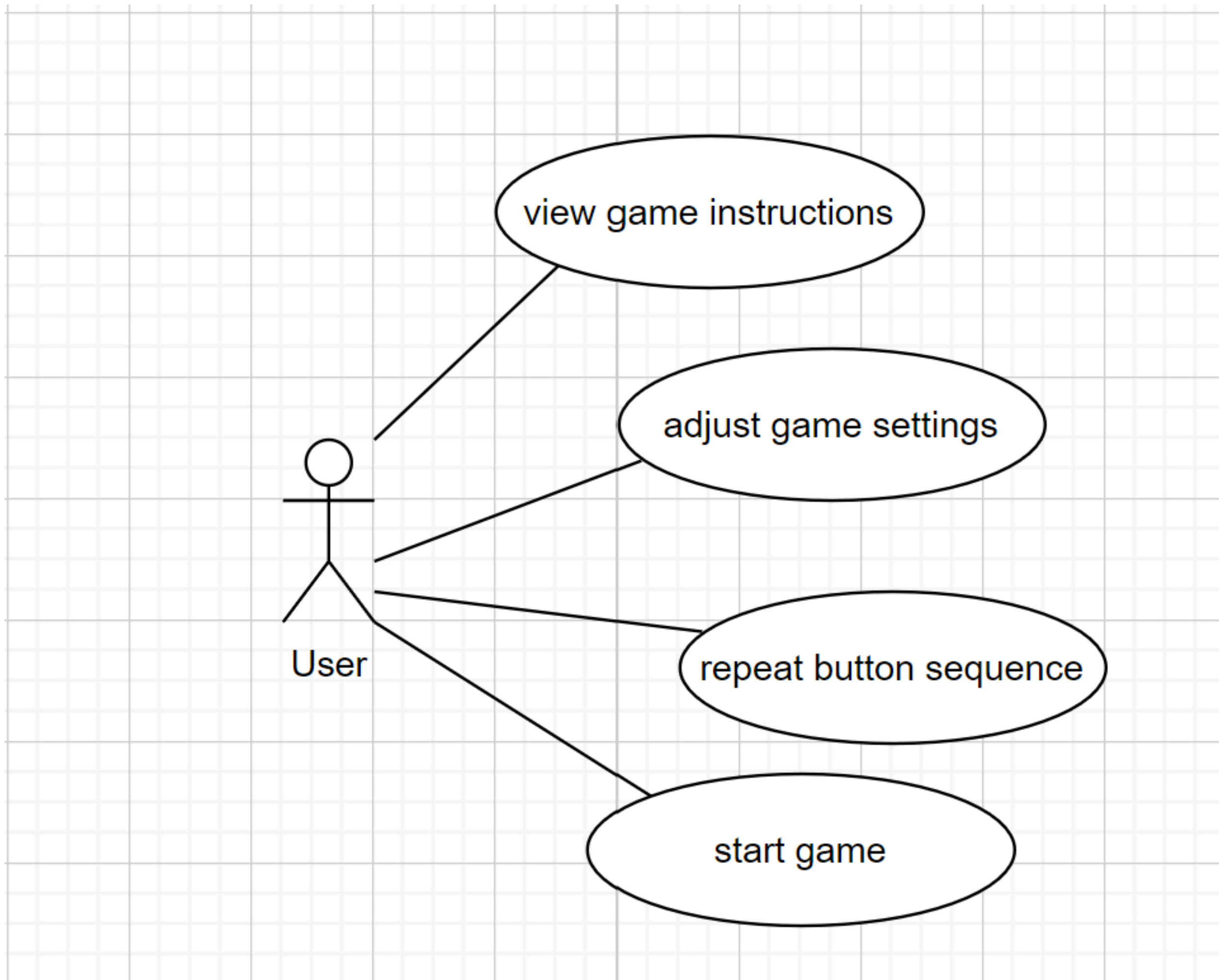
Help screen	This function tells the user how the game works
Inputs:	Input comes from when the player clicks on the help button on the main screen
Outputs:	The output of this function is the help screen being displayed.
Action:	The action of this function is to display text that helps the user understand how the actual game sequence works.
Pre-Condition	The user can be anywhere on the website to access the help screen- as the button is always displayed.
Post-Condition	The website will be on the help screen.
Side-effects	The game in session may be lost if the help screen is accessed in the middle of a game.

Sequence Creator/Checker	This will decide the sequence at random for the game and also store it in a data structure so the user's answers can be checked against it. Must also be able to add new steps in the sequence.
Inputs:	The inputs will be from the website and will be what the user has pressed on the website in correlation to the game. These inputs will be sent to the function and used to do the checking of answers.
Outputs:	The Outputs will be the pattern that the program has come up with as well as the same pattern in a data structure. The output will also be whether or not the user's input matches the pattern that the function has given. This will be sent back to the website in order to display the proper information.

Action:	The function will either create a new pattern for the game, or the function will check the users' answers against the pattern that it has made and return the answer to the website for display.
Pre-Condition: The player must have inputted a sequence for the system to check against its own created sequence.	This function will happen when a user inputs a sequence as well as when the user first opens the website to get the first sequence.
Post-Condition The system has either generated a new sequence or is returning if the user is correct or not	There will be a pattern that is sent back to the website, or there will be a result sent back to the website based on the users' answers.

Scoreboard	This will keep track of both the current score and also the high scores (locally) and give the values to the website to keep track of.
Inputs:	The input of this function will be whether or not the user got the pattern right. These inputs will be used to either return the current score or add to the current score.
Outputs:	The output of this function will be the current score and also the high score of the game. These will be sent to the website to display for the user.
Action:	The function will use the input that it gets to either return the current score if the user got the answer wrong or the score plus the points that the user just got.
Pre-Condition The user has played	Function Will be called anytime the user makes an answer to either update the score for the website or to return the current score to the user.
Post-Condition The users score	There will be a score sent to the website along with the high score.

Case Models:



View Game Instructions

Actors	User, application
Description	The user can access the instructions on how the game mechanics work
Stimulus	The user clicks the help button on the main menu screen
Response	The help screen is displayed
Comments	N/A

Adjust Game Settings

Actors	User, application
Description	The user can adjust various game settings in the settings screen
Stimulus	The user changes the setting(s) they want to change
Response	The appropriate setting(s) will be changed by the application
Comments	The current list of settings includes: changing of text size, changing of difficulty(amount of buttons) changing of sound

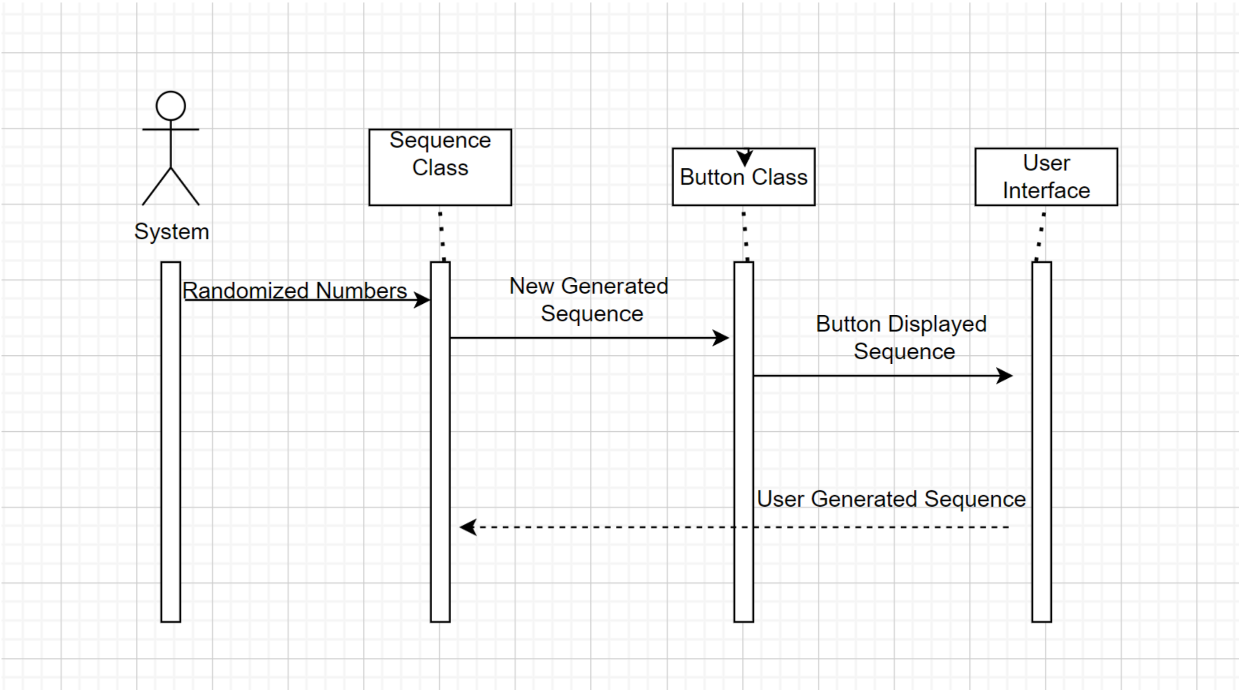
Repeat Button Sequence

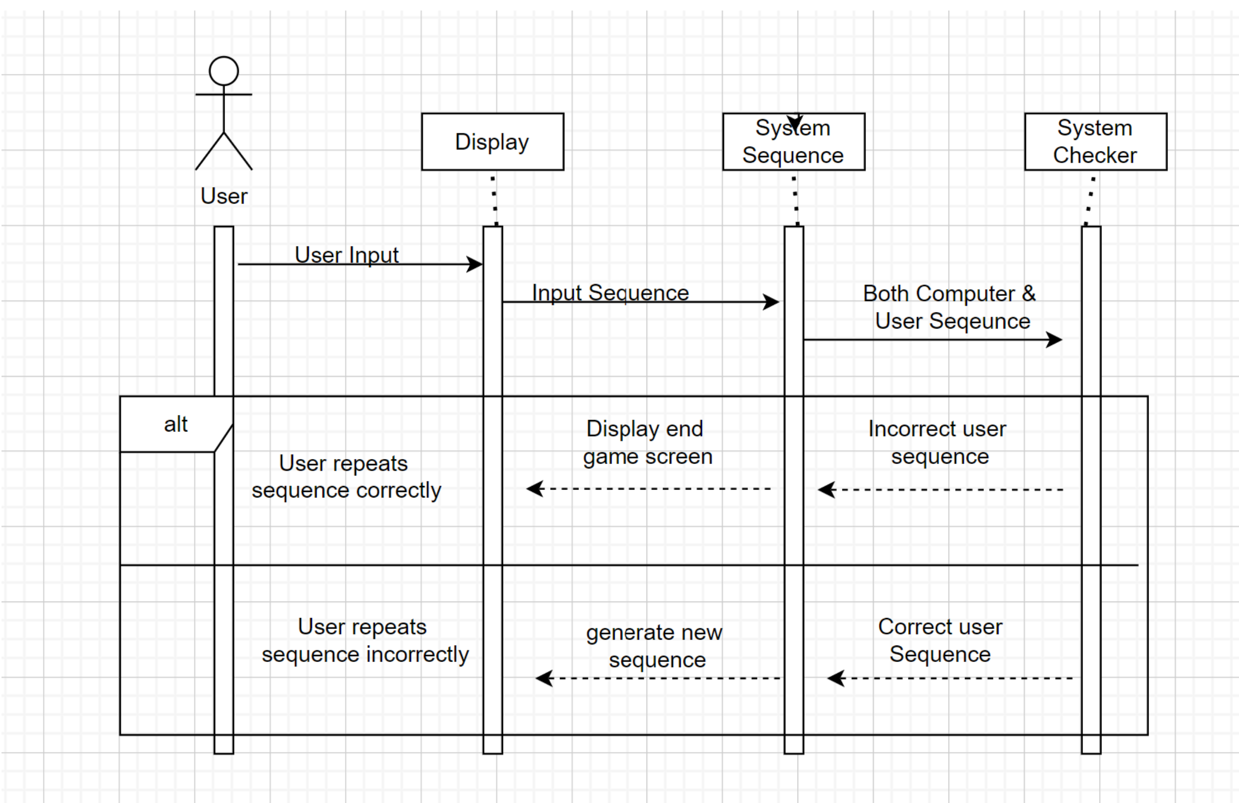
Actors	User, application
Description	The main function of this software is to get the user to repeat the button sequence that is generated by the code.
Stimulus	The user presses a sequence of buttons, trying to replicate the sequence the computer just showed.
Response	If the user matches the button sequence, then the same sequence with another button press at the end is added. If the user does not match the button sequence, then the application takes the user to the end game screen.
Comments	N/A

Start Game

Actors	User, application
Description	The user can start playing the main function of the game
Stimulus	The user presses the start game button on the main menu screen, or restart at the end game screen
Response	The application will start the sequence of buttons for the user to repeat back
Comments	N/A

Sequence Diagrams:





Glossary:

Users: These are the people, or sometimes another software/hardware, that are engaging with the game. These are the players who are watching the button sequence, and then trying to reproduce that sequence. In the context of this application, these are usually people with memory or mental impairment and older people.

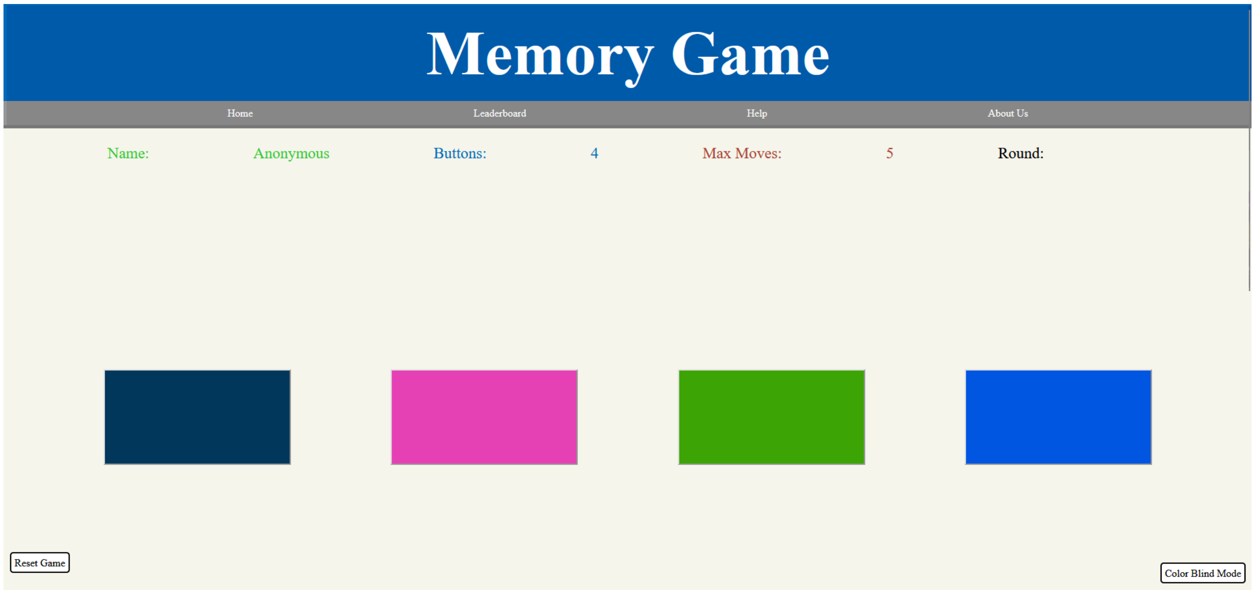
Colorblind Mode: A feature designed to accommodate users with color vision impairments. This mode adjusts the color palette of the main sequence buttons, which alters the game’s visual appearance. This helps enhance inclusivity and may help the user repeat the sequence, by having more visual cues to go off of.

Sequence: The computer-generated series of buttons presented in a specific order.

Tutorial: This is a feature of the application that is designed to explain the mechanics of the game and teach the player how to play.

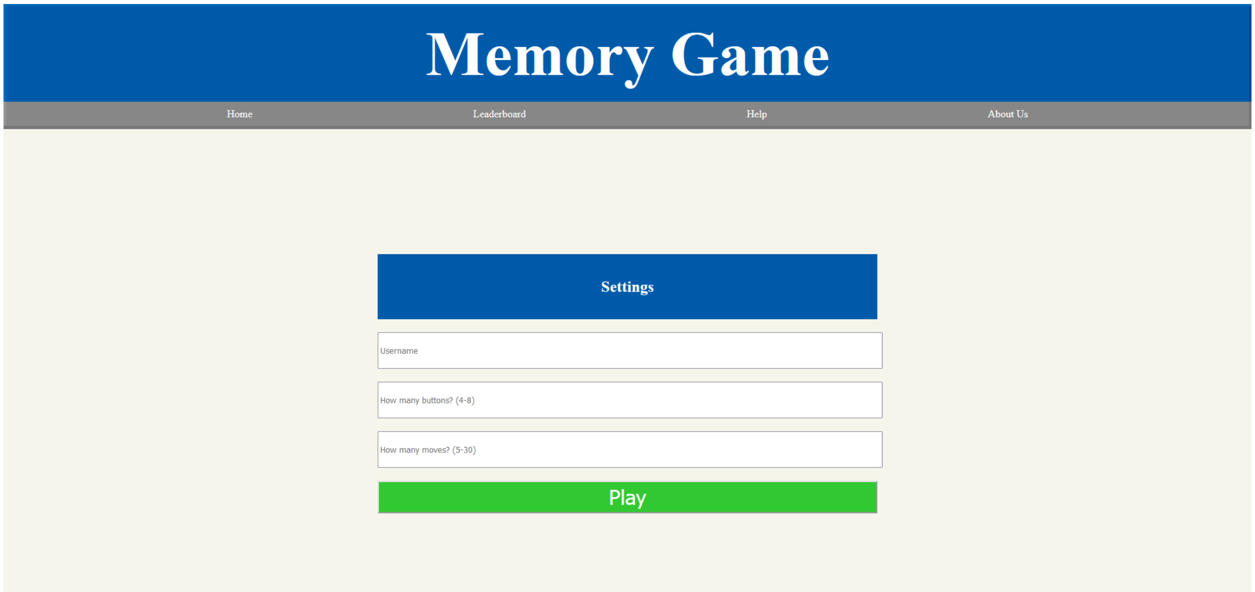
Score: This is a numerical representation of a player’s performance during a game session, it is based on the amount of buttons they have set multiplied by the amount of buttons they get right in a row.

Initial Screen Captures:

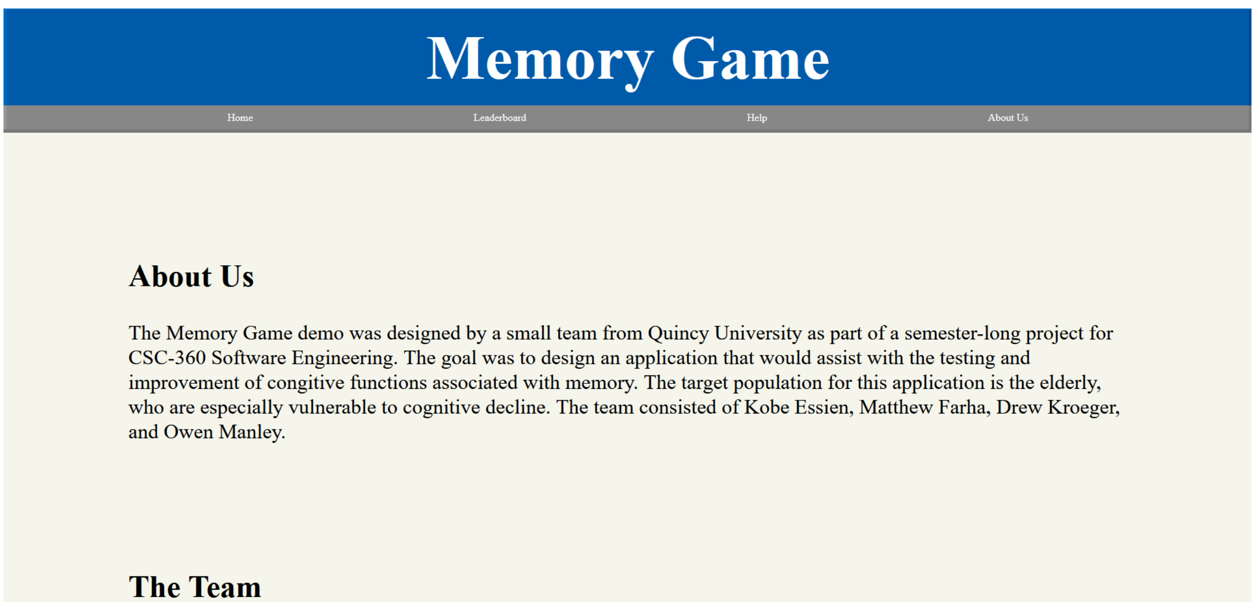


This is the main game sequence screen. The buttons light up in a given sequence, wait for player input, then light up in the same sequence and +1 move.

This is the help screen. It explains the basics of the game.



This is the main screen, it asks for the number of buttons the user wants(4-8), and the goal number of moves(5-30).



This is the about us screen.

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Contributions:

Drew Kroeger: Revised everything but system requirements, added more sources, added glossary, added user cases and descriptions, implemented screen captures, redid one system requirement

Kobe Essien- System Requirements Revisions, Sequence Diagrams
Owen Manley- helped with glossary, helped with requirements,
Matthew Farha- Worked on the game extensively, provided screen captures